Cardiovascular effects of consumption of black versus English walnuts.

English walnuts have been shown to decrease cardiovascular disease risk; however, black walnuts do not appear to have not been studied for their cardioprotective effects. The purpose of this study was to determine the effects of English versus black walnut consumption on blood lipids, body weight, fatty-acid composition of red blood cell (RBC) membranes, and endothelial function. Consumption of 30 g of English walnuts per day for 30 days, by 36 human participants, improved blood lipids; the effects of black walnuts were dependent on the participant's sex. Addition of either nut to the diet did not result in weight gain. The fatty-acid composition of RBC membranes was favorably affected by walnut consumption. RBC polyunsaturated fatty acids increased after consumption of either type of nut; however, eicosapentaenoic acid increased significantly more after English walnut consumption. Endothelial function of 6 unmedicated humans with hypercholesterolemia was maintained after consumption of English walnuts with a meal high in high saturated fats; however, consumption of black walnuts with the same meal did not maintain endothelial function. Overall, these results support the recommendation that consumption of 1 oz of English walnuts per day may decrease cardiovascular risk, but more research on black walnut consumption is necessary before an appropriate recommendation can be made.